

Arduino Microcontroller Concepts (CSE 4820)

(7th Semester, B.Tech)

ASSIGNMENT-1

(Submission deadline- 30.09.26)

1. What is the purpose of the setup() and loop() function?
2. What is the purpose of digitalWrite() and delay()? Explain with example.
3. Write the basic structure of an Arduino sketch.
4. What is variable scope? Explain local and global variables with example.
5. Write all the datatypes available in Arduino C++.
6. Write all the delay functions available in Arduino C++.
7. What is an Arduino String? Name any two string functions.
8. Write a program to blink an LED.
9. Write a Program to find the sum and average of 10 random numbers.
10. Explain and demonstrate the min() and max() functions. Write a code fragment that reads two sensor values and prints both the smaller and the larger of the two.
11. Two analog sensors provide readings through A0 and A1. Write an Arduino code fragment using abs() to calculate the difference between the readings and turn ON an LED if their difference is greater than 20.
12. Design an Arduino code fragment that processes a counter from 1 to 20 and turns an LED ON only when the counter reaches a multiple of 5. Use the modulo(%) operator and explain how it identifies the required values.
13. Write an Arduino sketch that gradually increases the brightness of an LED from 0 to 10 and then decreases it back to 0. Use ++ to increase the brightness and -- to decrease it. Display each brightness value on the Serial Monitor.
14. Why Does Overflow Occur? How we can Correct Overflow Handling. Explain through the example.
15. Write a program to generate 20 random numbers from 0 to 30.